

BIOLOGY 414
PRINCIPLES OF BIOLOGY II
Spring, 2018

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Join [UNHM Biology & Biotechnology Majors Facebook page](#), this is the bulletin board for the Biology and Biotechnology programs.

Participate in the Biology Club (announcement on FB page).

Description: Lecture and laboratory survey of the five kingdoms of life; physiology of cells, tissues, organs, and organ systems; evolution; human impact on the biosphere. Required for students majoring in the life sciences. Cannot be taken for credit after BIOL 412 or equivalent.

Objectives: Upon completion of this course you should have acquired a basic understanding of evolution, plant organismal biology, animal organismal biology, and ecology.

Prerequisites: None

Textbooks: Campbell, et al. 2014. 2nd Edition. *Biology in Focus*. Pearson. ISBN 9780321962751 (hardcover), **or** loose leaf **or** eBook.

Symbiosis. 2010. Principles of Biology Lab Manual. Pearson Learning Solutions. ISBN 978-0-536-37730-2.

A lab coat is required and may be purchased at the UNH-M bookstore.

Grading:	3 lecture exams @ 100 points.....	300 pts
	1 research project	100 pts
	Best 10 of 12 quizzes taken @ 10 points each.....	100 pts
	3 laboratory exams @ 20 points.....	60 pts
	8 laboratory quizzes @ 5 points.....	<u>40</u> pts
	Total Points Available.....	600 pts

All exams must be taken when scheduled. Make-up exams will be permitted only for excused absences (e.g., confirmed illness or family emergency) or if arranged in advance. Each exam only covers material since the previous exam; i.e., the second and third exams are not cumulative. Quizzes will consist of 10 multiple choice questions covering material from the previous class. They must be taken on Blackboard and must be taken by the due date. **The lowest 2 grades for quizzes that you take will be dropped** (that means if you don't take a quiz, the zero counts). There are no makeups of quizzes. Your course grade will be determined based on the percentage of points earned out of the possible 600. 90% and above will earn A- or better; 80% and above will earn B- or better; 70% and above will earn C- or better; 60% and above will earn D- or better.

Academic Honesty

The Biology Program at UNH Manchester will strictly adhere to the University policy on academic honesty, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (<http://www.unh.edu/vpsas/handbook/academic-honesty>). By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work - in whole or in part - as your own; collaborating on answers for individual assignments, or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report.

Cheating is mainly concerned with copying on exams or in lab, bringing crib notes into an exam or referring to notes or the textbook or any other source such as a programmable calculator, tablet, or cell phone during an exam. All electronic devices must be turned off and put away for the duration of the exam. Ear buds are not permitted.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

Expectations and Attendance:

Attendance in both lecture and laboratory is required. Excessive absences may result in a reduction of your grade. It is expected you spend a considerable amount of time outside of class reading, reviewing and studying class material. For every hour spent in lecture you should spend three hours studying. It is expected you will attend study group or go to a tutor once per week.

Center for Academic Enrichment (CAE)

The **Center for Academic Enrichment** is a resource for all students enrolled in **UNH Manchester** courses. Our mission is to help under-prepared students prepare, prepared students advance, and advanced students excel. Trained peer tutors and professional staff provide free assistance with reading and writing in all courses, as well as content in all math and science courses. Support for other content areas is dependent on the background and competencies of current peer tutor staff. ***All enrolled students are entitled to 1 hour of free one-to-one tutoring per week, per course.*** In addition, peer and professional tutors may run study groups for your class and provide drop-in math tutoring. Please visit us online at manchester.unh.edu/student/center-academic-enrichment to learn more about our services or to make an appointment with a tutor. There will be study group every Tuesday before lecture. You are strongly advised to attend with material you want to review. There is a direct correlation between students who attend study group and students who earn high grades in this class. Invest your time in this opportunity. The room assignment for study group will be announced after the first week of class.

ADA Statement:

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must register with the Disability Services Office. The Disability Services Coordinator at UNHM is Jenessa Zurek. Jenessa can be contacted at (603) 641-4170, jenessa.zurek@unh.edu or in person in the

Student Services Suite, Room (#410H).

Human Disease Research Project:

1. Choose a human disease and submit it via the assignment tool in Canvas. Do not attach a file just write the disease in the textbox. I will provide feedback in Canvas
2. Read an article in Access Science or Encyclopedia Britannica to gain general knowledge about your topic.
3. Pose a thoughtful question about your disease. Submit both the URL of the article and your question. Do not attach a file just write the disease in the textbox. I will provide feedback in Bb.
4. Find three journal articles using Academic Search Complete (through UNHM Library) that can help you answer the question you posed. Put your references in APA format. Write a one paragraph (100 words) summary of each article. Upload a Word or .rtf file for this assignment.
5. Write a 3-5 page paper of your research paper where you attempt to answer your question. This paper is based on your three journal articles and your Access Science/Britannica article. Your in text citations should be in APA format. Include your list of four references.
6. After seeing the tutor with your paper, make all revisions and hand in the final version of your paper. Late papers will have points deducted.

All assignments (except for final version of paper) must be turned in by 11:59 pm on the due date via the assignment tool in Blackboard. Late assignments will receive 0 points.

<u>Assignment</u>	<u>Due date</u>	<u>Points</u>
Submit name of disease	February 7	5
Question posed about disease; URL of Access Science article	February 14	5
Annotated bibliography in APA format	March 7	10
Meet with tutor and submit signed tutor sheet via Canvas	April 11	10
Final revised paper via Canvas	April 18	70

Lecture Schedule:

<u>Date</u>	<u>Topics</u>	<u>Reading Chapter#s</u>
Jan 24	1. Evolution & Natural Selection	1, 19
Jan. 31	2. Population Genetics & Speciation	21, 22
Feb 7	3. History, Classification, & Diversity of life	20,23,24,27
Feb 14	4. Plants I: Structure, Growth, & Transport	28-29
Feb 21	5. Plants II: Reproduction and Development	30
Feb 28	EXAM I	
Mar 7	6. Animal Systems I: Tissues, Homeostasis, Digestion & Nutrition	32, 33
Mar 14	SPRING BREAK	
Mar 21	7. Animal Systems II: Circulation & Gas Exchange	34
Mar 28	8. Animal Systems III: The Immune System	35
Apr 4	9. Animal Systems IV: Salt and Water Balance; Nitrogen Excretion	32
Apr 11	EXAM II	
Apr 18	10. Animal Systems V: Reproduction & Development	36
Apr 25	11. Animal Systems VI: Nervous & Sensory (online lecture)	37, 38
May 2	12. Ecology: Ecosystems, Communities, & Populations	40-43
May 9	EXAM III	

2018 BIOL 414 LAB SCHEDULE

JAN 31, Feb 1, 2	Introduction/Overview/Lab Safety/ Effectiveness of Hand Washing 1	page 289
FEB 7, 8, 9	Effectiveness of Hand Washing Part 2	page 289
FEB 14, 15, 16	Domain Bacteria + Domain Archae	page 1
FEB 21, 22, 23	Kingdom Protista and Fungi	page 35
FEB 28, MAR 1, 2	Plant Anatomy	page 73
MAR 7, 8, 9	Plant Diversity I + II	page 107+133
MAR 14, 15, 16	SPRING BREAK	
MAR 21, 22, 23	LAB EXAM I	
MAR 28, 29, 30	Animal Diversity I	page 161
APRIL 4, 5, 6	Animal Diversity II	page 183
APRIL 11, 12, 13	Vertebrate Anatomy I	page 217
APRIL 18, 19, 20	Vertebrate Anatomy II	page 243
APRIL 25, 26, 27	Vertebrate anatomy III	page 265
May 2, 3, 4	LAB EXAM II	